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UNIVERSITY OF CAMBRIDGE INTERNATIONAL EXAMINATIONS
General Certificate of Education Ordinary Level

CANDIDATE NAME

CENTRE NUMBER

CANDIDATE NUMBER

4033705314

BIOLOGY **5090/33**

Paper 3 Practical Test **May/June 2010**

1 hour 15 minutes

Candidates answer on the Question Paper.

Additional Materials: As specified in the Confidential Instructions.

READ THESE INSTRUCTIONS FIRST

Write your Centre number, candidate number and name on all the work you hand in.
Write in dark blue or black pen.
You may use a pencil for any diagrams, graphs or rough working.
Do not use staples, paper clips, highlighters, glue or correction fluid.
DO **NOT** WRITE IN ANY BARCODES.

Answer **both** questions.
At the end of the examination, fasten all your work securely together.
The number of marks is given in brackets [ ] at the end of each question or part question.

For Examiner's Use	
1	
2	
Total	

This document consists of **8** printed pages.

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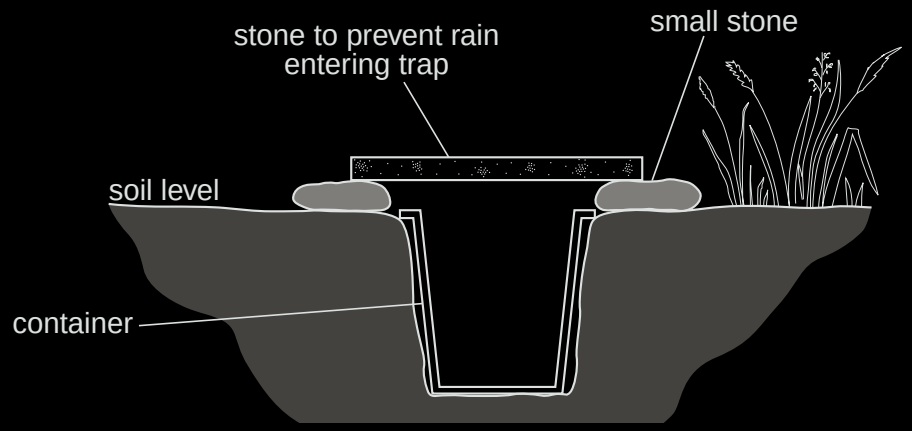


Fig. 2.1

A number of pitfall traps are placed in the habitat and left for a period of time.
The containers in the pitfall traps are collected, any beetles in them are counted and recorded **[A]**, marked and then released back into the same habitat.

The containers in the pitfall traps are replaced after a day and left for the same period of time as before.
Again the containers in the traps are collected and the number of beetles counted and recorded **[B]**.
The number of beetles with marks in this second group are counted and recorded **[C]**.

The formula $\frac{\mathbf{A} \times \mathbf{B}}{\mathbf{C}}$ can then be used to given an estimate of the total number of beetles in the habitat.

(c) (i) You are provided with a bag of beans to represent the beetles, labelled **S5**.

- Remove 12 beans. These represent the beetles in the first sample **[A]** and this figure has been recorded already in Table 2.1.
- Mark each of these 12 beans with a visible dot using the marker pen provided.
- Put these marked beans back into the bag. Shake the bag gently to mix these with the other beans.
- Remove a handful of beans and place on the bench in front of you and count them.
- Record the total number of beans in this handful in Table 2.1 in column **B**.
- Count and record separately the number of beans in this handful that have been marked and record this in column **C** in Table 2.1.

Table 2.1

A	B	C
12		

[2]

